2023 45th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC) | 979-8-3503-2447-1/23/$31.00 ©2023 IEEE | DOI: 10.1109/EMBC40787.2023.10340498

Integrating Machine Learning with Biomedical Signal Processing and Systems Analysis: An Applications-based Course

Patjanaporn Chalacheva, *Member, IEEE*, and Michael C.K. Khoo, *Fellow, IEEE*

***Abstract*— The growing importance of data analytics in biomedicine is increasingly becoming recognized in biomedical engineering curricula through the introduction of machine learning classes that generally run in parallel to, but separately from, more traditional engineering courses, such as signal and systems analysis. We propose a new approach that systematically integrates signal processing and systems analysis with key techniques in machine learning. In the proposed course, the student obtains hands-on experience in applying algorithms that can be applied to practical problems of physiological signal conditioning, analysis and interpretation. This is achieved by exposing the student to a sequence of 4 applications-based modules that represent different biomedical engineering problems: human activity recognition from wearable devices, epileptic seizure detection, quantification of dynamic respiratory-cardiac coupling in humans under different conditions, and detection of sleep apnea episodes from heart rate variability data. Within each module, the student gains the experience of working with the data in question "from the ground up". We also introduce a general plan for assessment of student learning, and discuss the expected outcomes and limitations of this integrative approach to teaching.**

***Clinical Relevance*— The proposed course is targeted at biomedical engineering students at the senior undergraduate or first-year graduate level who are interested in learning how to analyze physiological signals. The course would also be suitable for clinician-scientists who have prior training in statistics with some exposure to engineering mathematics.**

I. Introduction

Over the past decade or so, the field now known as "Data Analytics" has grown exponentially along with the ubiquity of computing power and the development of new algorithms and applications in all areas of knowledge, including biomedicine. One major methodological focus comes under the name of "Machine Learning", and although it is considered a relatively new field, the use of advanced mathematical and computational methods for the analysis of biomedical data and modeling of physiological systems dates back to the neuron model first introduced Pitts and McCullogh in 1943 [1]. Today's examples of machine learning applications in the biomedical engineering domain include medical signal/image recognition and classification [2,3], clinical decision support [4] and treatment monitoring [5]. Successful application of machine learning algorithms to data and the explosion of career opportunities in data analytics have stimulated a great deal of interest among engineering students, and consequently, an increasing number of engineering programs have started to offer "Introduction to Machine Learning" courses. However, these introductory courses to machine learning tend to focus on teaching students various supervised and unsupervised machine learning algorithms and the application of these algorithms to well-curated, domain-specific datasets [6,7]. The focus is generally on the principles and derivation of the algorithms themselves, with much emphasis on selecting an algorithm that attains the highest accuracy in prediction. Moreover, in many courses, the datasets have already undergone prior signal pre-processing, feature extraction and/or data reduction. This contrasts with the actual process of analyzing real biomedical datasets which are generally noisy and contain measurement artifacts or loss of signal. We therefore believe that it is important for the student to "start from the ground up": develop an understanding of how the datasets were measured, be able to discern what aspects represent physiological fluctuations and what constitute "noise", and thus have a good "sense" of the data before applying machine learning algorithms. This is particularly important if a key goal, aside from prediction accuracy, is to gain insight into the underlying physiological mechanisms that generate the measured signals.

To address the aforementioned shortcomings, this work proposes a new biomedical engineering course that carefully integrates the teaching of signal processing and systems analysis with key machine learning concepts and algorithms by adopting a systematic approach that exposes the student to a sequence of 4 applications-based modules.

II. Course design

This course introduces the basic principles and algorithms employed in the processing, analysis and interpretation of cardiovascular, respiratory and electrophysiological signals acquired in humans through medical monitoring systems and wearable devices. Students will learn traditional filtering, spectral analyses, and linear and nonlinear dynamical analyses. These techniques can be used to extract compact, yet meaningful, information relevant for clinical diagnostic and prediction. In addition, students will learn methods for dimensionality reduction and supervised and unsupervised learning. We will approach these topics by studying different real-world biomedical engineering applications.

A. Target students

This course targets primarily first-year graduate students and advanced undergraduate students in Biomedical Engineering or related disciplines. Students should have background in basic engineering mathematics such as linear

P. Chalacheva is with Biomedical Engineering, Carnegie Mellon University, Pittsburgh, PA 15206 USA (phone: +1 (412) 268-6282; e-mail: pchalach@andrew.cmu.edu).

M.C.K. Khoo is with the Alfred E. Mann Department of Biomedical Engineering, University of Southern California, Los Angeles, CA 90089 USA (phone: +1 (213) 740-0347; email: khoo@usc.edu).

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algebra, power series and basic probability. Student should also have experience in programming in MATLAB and/or Python.

B. Course objectives

Upon successful completion of this course, students should be able to: 1) understand and apply the basic principles underlying time- and frequency-domain approaches to discrete-time analysis of biomedical signals; 2) understand the unique characteristics of various kinds of physiological time-series, including beat-to-beat signals, and how to deal with the practical issues related to the detection, acquisition filtering, and physiological interpretation of these signals; 3) apply methods of analysis that quantitatively characterize the extent of complexity and nonlinearity in biomedical signals; 4) appreciate the connection between the "traditional" engineering data-driven methodologies with the "new" machine learning algorithms; and 5) detect and classify dynamic patterns in physiological signals using machine learning approaches.

C. Course design

This course comprises four modules for different biomedical engineering problems. Each module covers both signal processing and machine learning methods that are applicable to the selected problem.

1) Human activity recognition

With increasingly available wearable technologies and devices, consumers can routinely use sensors for measuring activity unobtrusively and continuously. Activity recognition by wearable devices is potentially useful for health and wellness management. In this module, students will learn how to classify different patterns in the accelerometer signals into different activities from publicly available dataset such as the Human Recognition Using Smartphones dataset, UCI Machine Learning Repository [8].

In the first module, students will be exposed to typical techniques for data pre-processing and the general workflow of machine learning model development. The module starts with the instrument principles (in this case, physical principles of accelerometer and the measured acceleration), data acquisition and sampling theory (Nyquist rate and aliasing). Next, students will learn techniques for data cleaning such as interpolation and filtering and convolution, and how to extract compact characteristics from signals such as average, variance and standard deviation, root mean square, etc. These extracted characteristics can be used to represent different patterns in the accelerometer signals in different activities, and can be used as the inputs or features in subsequent machine learning models.

Before introducing machine learning algorithms, students need to first understand different phases in the implementation of a machine learning model. This involves building datasets, model training, testing, refinement and evaluation. In this first module, students will learn two simple and easily interpretable machine learning algorithms for activity recognition: K-nearest neighbors [9] and decision trees [10]. Additionally, students will learn how to construct and interpret confusion matrix, evaluation metrics and receiver operating characteristic (ROC) curves.

2) Epileptic seizure detection

Electroencephalography (EEG) plays an important role in detecting epilepsy. An epileptiform activity in EEG signals can manifest as spikes, sharp waves, or spike-and-wave complexes – all of which alter the frequency contents in EEG signals. As such, students will learn frequency-domain analyses as a means of extracting different patterns in the EEG signals in the presence or absence of seizure activities. One of the possible data sources for this module is the Epileptic Seizure Recognition dataset, UCI Machine Learning Repository [11].

Similar to the previous module, this module begins with background information on the principles of EEG signals and how EEG signals may change during epileptic seizure. Next, students will be introduced to frequency analyses e.g., Fourier transform and power spectrum analysis. It should be evident to students that EEG segments with spikes, which may be reflective of epileptic seizure, contain different frequency composition compared to no-event EEG segments.

Students will follow the general machine learning model development workflow discussed in the previous module. In addition, we will introduce cross-validation for model refinement and evaluation in this module. Two new machine learning algorithms introduced in this module are: support vector machines [12] and random forest models [13], an ensemble learning method, which is an extension on the decision tree algorithm introduced in the first module. Students will also learn principal component analysis, a data reduction technique that can be useful for this and other applications [14].

3) Respiratory-cardiac coupling in sleep apnea – time-series forecasting

This module focuses on modeling and quantifying relationships between signals. The students begin this module by learning about the physiology involved in respiratory-cardiac coupling and sleep apnea. Students will also learn how to extract beat-to-beat heart rate/period from electrocardiogram (ECG) signal. Potential data sources for this module are from Sleep Data – National Sleep Research Resources [15].

Techniques used to quantify dynamical relationship between signals will be introduced such as cross-correlation, impulse response estimation (via one-step inversion of the observations matrix, as well as iteratively using gradient descent), and cascade models such as the linear dynamic system to static nonlinearity ("Wiener") cascade model [16]. The identified relationship from the linear or nonlinear model can then be used to predict the output time-series, given the input time-series.

The new machine learning algorithms introduced in this module are perceptron and multilayer perceptron (MLP) [17], where both models will be used for time-series forecasting i.e., predicting the next time-step of the heart rate/period given past and current respiratory input signal. MLP structure, forward propagation and backpropagation will be discussed. Additionally, we will draw direct comparison between the "traditional engineering" methods for dynamic modeling introduced earlier in this module with perceptron and MLP models with linear activation function.

4) Sleep apnea classification using heart rate variability

This module is a continuation of the previous module but shifts the focus from predicting time-series to predicting class labels (e.g., using only heart rate data, can we tell whether a data segment contains a sleep apnea event or not?). Logistic regression will be introduced. Perceptron and MLP models will be revisited but this time with nonlinear activation, which allows thresholding the predicted probabilities, and subsequently classify data into classes.

Convolutional neural networks (CNN) [18] will also be introduced in this module. CNN model structure and basic building blocks in CNN models (e.g., convolutional layers, pooling layers, and dropout layers) and their operations on data will be discussed. In addition, we will draw direct comparison between 1D-CNN and Volterra models; the latter class of models has been used for nonlinear dynamic modeling in bioengineering applications [19]. A Volterra model consists of filter banks, analogous to convolutional layers in CNN model, followed by nonlinear function that produces an output, analogous to nonlinear activation function. Students can also visualize and compare the estimated weights of filters in convolutional layers with the estimated coefficients of the Volterra model filter banks. The course design: sequence and topics covered are summarized in Fig. 1 below.

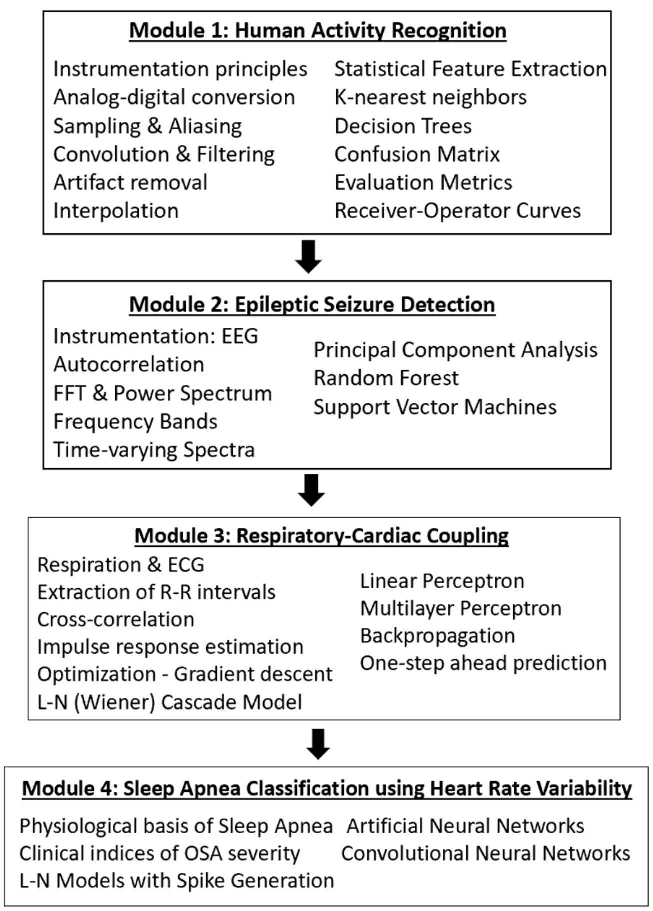

Figure 1. Course design summary, showing the sequence of applications and topics covered. "OSA" in Module 4 stands for obstructive sleep apnea.

III. Assessment Plan

Each module will be accompanied by a homework assignment consisting of quantitative problems related to the topics in that module. Assignments in Module 1 and 2, which cover basic techniques in signal processing and simple machine learning algorithms, will be structured such that there are two parts to each assignment. The first part guides the student through the general workflow in data analysis and is aimed at solidifying the student's "know how" and "know why". The "know how" can be assessed by asking students to perform specific tasks e.g., filtering, calculating power spectrum, and programming. The "know why" can be assessed by asking questions related to the fundamental concepts learned in class e.g., determine the minimum frequency content that can be derived from a data segment of X duration. The second part of the assignment will be more open-ended, allowing the student to explore how changes in data processing steps and/or machine learning model parameters or structures can affect the model output and its interpretation. These assignments primarily assess lower level of learning in the Bloom's taxonomy [20]: remembering, understanding, applying, and to a limited degree, analyzing. These assignments are aligned with course objectives 1, 2 and 5 (see Section II.B. Course objectives).

Upon completion of Module 2, students should already have a good grasp of the general workflow of data analysis. As such, assignments for Modules 3 and 4 will be relatively open-ended. While students will work on the same application – respiratory-cardiac coupling in sleep apnea, they can choose to work on different datasets from the recommended Sleep Data – National Sleep Research Resources. They will have freedom to develop their own models for estimating the said mechanism. The only constraint is that students must employ at least one "traditional" engineering approach and at least one machine learning approach. Additionally, students must provide justification on their design decisions. This type of assignments allows assessment of higher level of learning in the Bloom's taxonomy: analyzing, evaluating and creating. Since students may have arrived at different solutions to the assignments, students will be graded based on the appropriateness of the selected methods with sound justification, their ability to correctly execute the chosen methods, the ability to compare and interpret the model outputs from the "traditional" engineering and machine learning approaches, and the ability to draw the final conclusion about the respiratory-cardiac coupling and heart rate variability in sleep apnea. These assignments are aligned with all five course objectives (see Section II.B. Course objectives).

IV. Discussion

With growing interest in data analytics and machine learning applications in biomedicine, many engineering programs have begun to offer an introductory course to machine learning as a separate course in addition to the "traditional engineering" courses. However, as instructors who have taught biomedical signal processing as a separate course from introductory machine learning for biomedical engineers at the senior undergraduate and first-year graduate levels, we have observed that students often fail to make the connection between the signal analysis and machine learning model development aspects of the problem whereas these two facets are actually intertwined.

This proposed course is designed to integrate signal processing, systems analysis and machine learning and present these concepts to students a sequence of applications-based modules. We expect that students will gain a better understanding of the entire data analysis process from the ground up. They will understand how the design choice for data conditioning can affect the extracted features and subsequently the performance, interpretability and generalizability of the developed machine learning model. Additionally, putting equal emphasis on both data conditioning and machine learning algorithms should alleviate the tendency for students to focus their efforts on refining only the machine learning models to attain the highest accuracy – a common drawback in introductory machine learning courses that pay little attention to the nature and quality of the datasets being analyzed.

However, this proposed course is not without its limitations. We recognize that integrating signal processing and machine learning into one course will result in having to cover a large number of topics in one semester. As a result, we cannot introduce students to as broad a variety of machine learning algorithms as one might be able to cover in a generic Introduction to Machine Learning course. To overcome this limitation, one solution could be to expand the proposed course into a two-semester offering that includes additional types of data (e.g., medical images, data containing discrete events such as neural spikes) and machine learning algorithms. We also recognize that learning signal processing and machine learning concepts, in general, can be challenging and demanding. However, our premise is that introducing these concepts from the application perspective with hands-on exercises will increase student motivation and facilitate the learning process. What we have proposed is meant to initiate the discussion among interested instructors about "best practices" in the ongoing quest to introduce machine learning to biomedical engineering students.

V. Conclusion

This work outlines a proposed course that introduces machine learning concepts with integrated signal processing and systems analysis from real-world biomedical application perspectives. Our ultimate goal is to teach students to recognize the well-known adage of "garbage in garbage out", and that a successful development of machine learning model takes into consideration not only model performance but also the data that are fed into the model. The proposed course may be considered a vertical integration approach that teaches students to begin the data analysis workflow from the level of raw data (post-acquisition) to the development of high-level models that may employ both classical dynamic systems approaches along with contemporary machine learning algorithms.

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